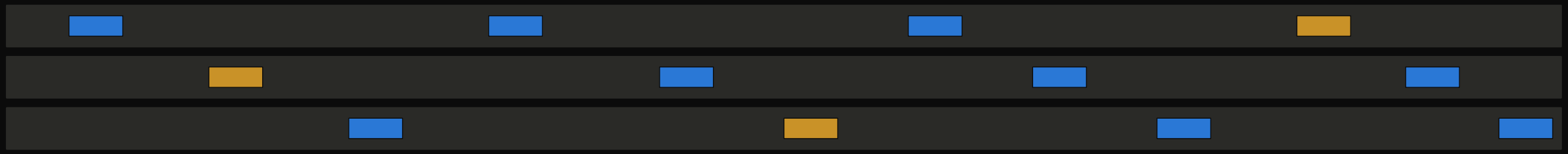
SELF-SOVEREIGN IDENTITY FOR CONNECTED VEHICLES

Every V2V message signed, every receiver verifying: the identity layer, shown

Nikhil Prakash · MASc, Electrical and Computer Engineering, UBC



blue = SSI / MOBI VID ochre = PKI

FOUR QUESTIONS

RQ1

Cost

What does a vehicle identity cost on chain, standard by standard?

RQ2

Security

Which architecture rejects the V2X attacks of the threat model?

RQ3

Compliance

Can a blockchain identity layer meet W3C DID and VC and MOBI VID?

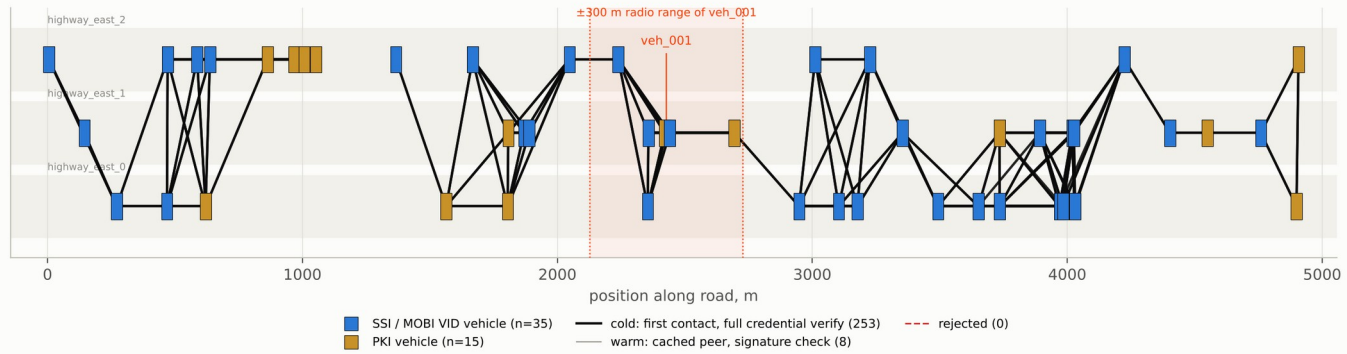
RQ4

Real time

Does credential verification fit the 100 ms V2V safety budget?

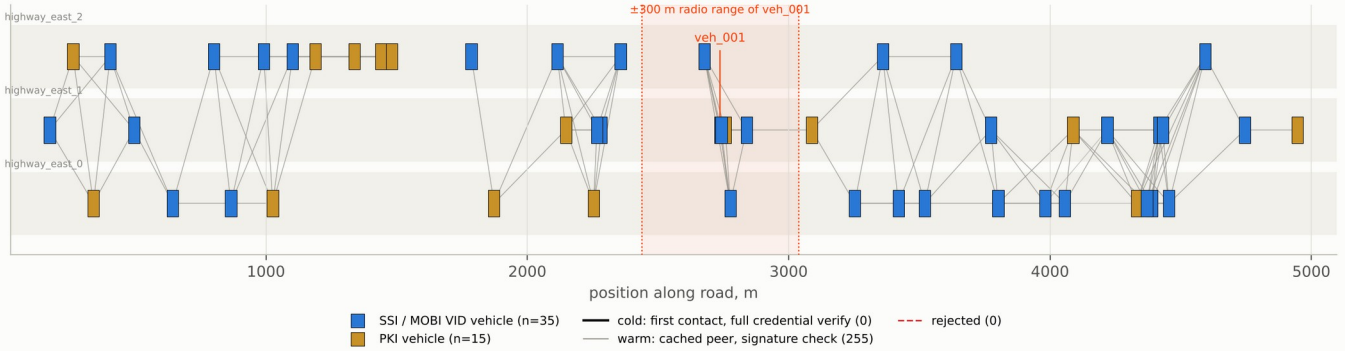
THE ROAD, 100 MS AT A TIME

THE ROAD AT T = 0.0 S
One 100 ms step: 51 signed BSM broadcasts, 261 receiver-side verifications, every one a real cryptographic operation.



trace sim_50v_20s_seed6.jsonl.gz · mode simulate · seed 6 · 20 s · 10 Hz BSM · 300 m radius · ≤8 rx/broadcast · commit 61d7157 · no radio channel, no MAC

THE ROAD AT T = 15.0 S
One 100 ms step: 51 signed BSM broadcasts, 255 receiver-side verifications, every one a real cryptographic operation.



trace sim_50v_20s_seed6.jsonl.gz · mode simulate · seed 6 · 20 s · 10 Hz BSM · 300 m radius · ≤8 rx/broadcast · commit 61d7157 · no radio channel, no MAC

t = 0 s

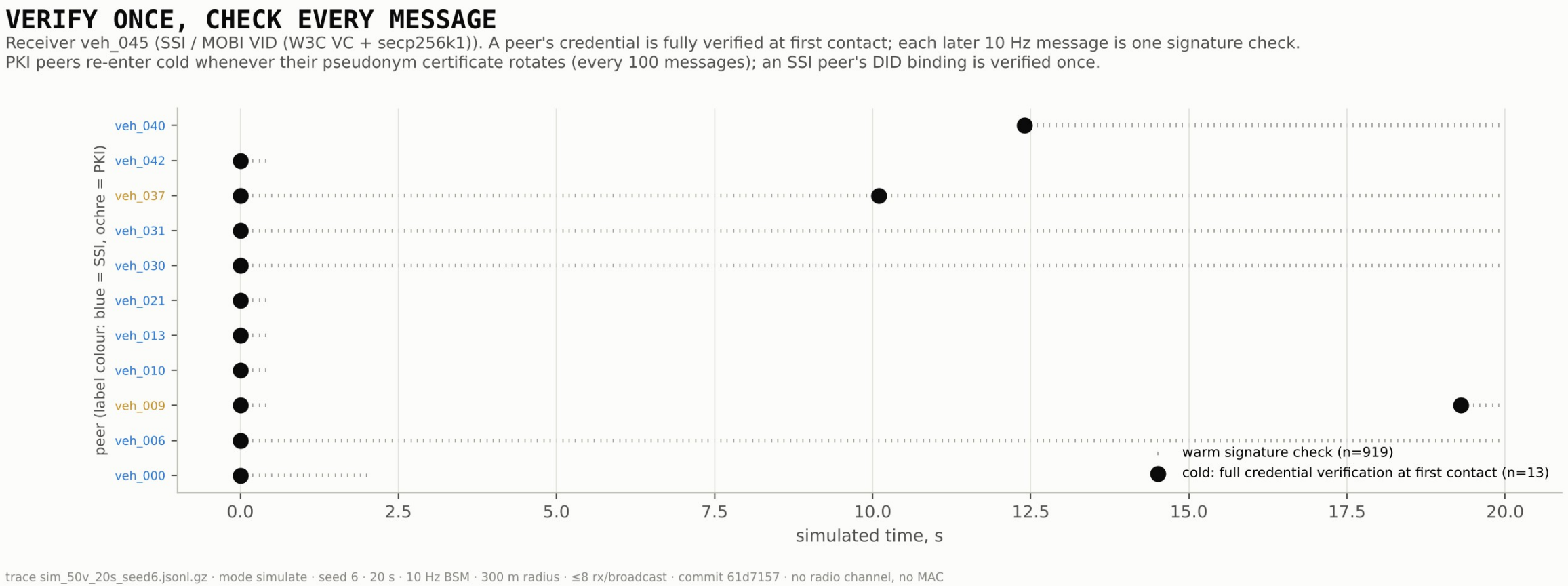
Every neighbour is a first contact: links are cold, full credential verification.

t = 15 s

Steady state: one signature check against a cached key per message.

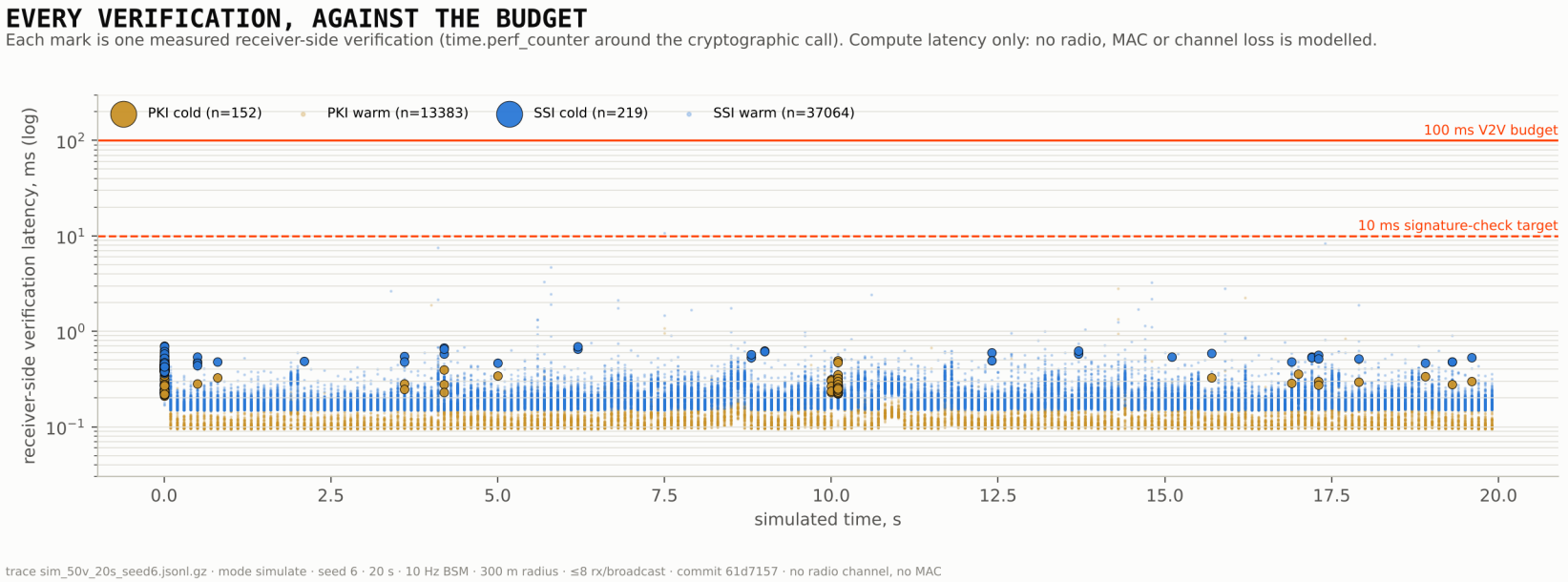
50 vehicles, 10 Hz, seed 6 of the registered N = 30 set; mock mobility, no radio channel.

VERIFY ONCE, CHECK EVERY MESSAGE



PKI peers re-enter cold at every pseudonym-certificate rotation (every 100 messages); an SSI peer's DID binding is verified once per encounter.

EVERY VERIFICATION, AGAINST THE BUDGET



0.142 ms

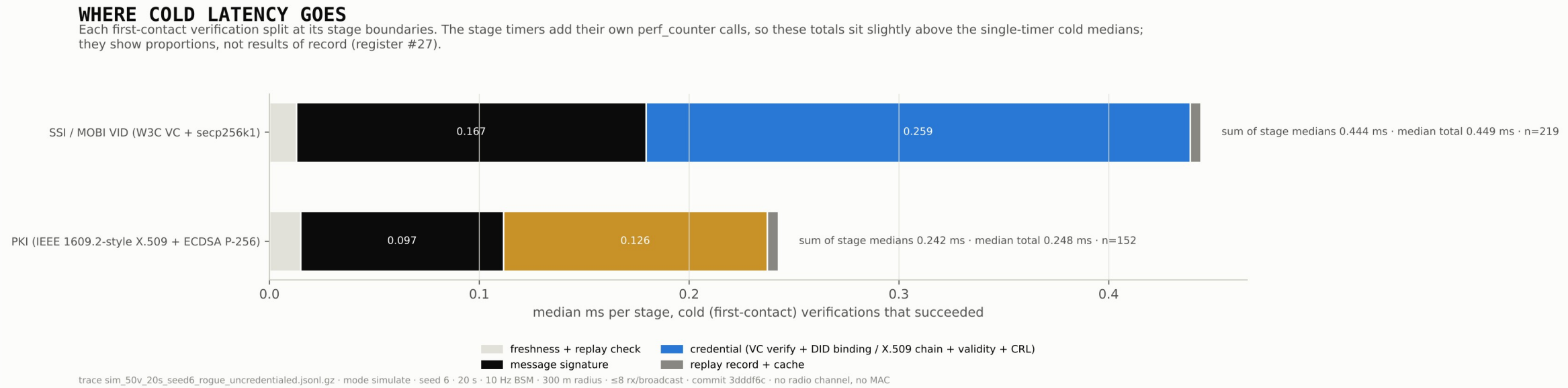
SSI warm verify, median of 30 run-medians (register #27)

100 ms

SAE J2945/1-derived V2V budget

Compute latency only: no radio, MAC or channel loss is modelled.

WHERE COLD LATENCY GOES



0.259 ms

SSI credential step: W3C VC verify + DID binding

0.126 ms

PKI credential step: X.509 chain + validity + CRL

Measured with stage timers on, so these are proportions; results of record stay #27.

SEVEN ATTACKS, SEVEN REJECTIONS

ATTACK BOARD

7/7 injected bad messages rejected at t = 19.9 s. Pass/fail is computed by the receiver's real verification path, not asserted.

Tampered PKI BSM speed field altered after signing	REJECTED
Tampered SSI BSM position altered after signing	REJECTED
Stale PKI BSM genuine signature, timestamp 5 s old	REJECTED
Stale SSI BSM genuine signature, timestamp 5 s old	REJECTED
Replayed PKI BSM accepted once, delivered again	REJECTED
Replayed SSI BSM accepted once, delivered again	REJECTED
Uncredentialed sender valid key, DID, no credential	REJECTED

trace sim_50v_20s_seed6.jsonl.gz · mode simulate · seed 6 · 20 s · 10 Hz BSM · 300 m radius · ≤8 rx/broadcast · commit 61d7157 · no radio channel, no MAC

Injected after every run, in both populations:

tampered, stale and replayed BSMs, and a sender with a key and a DID but no credential.

Pass or fail is computed by the receiver's real verification path, not asserted.

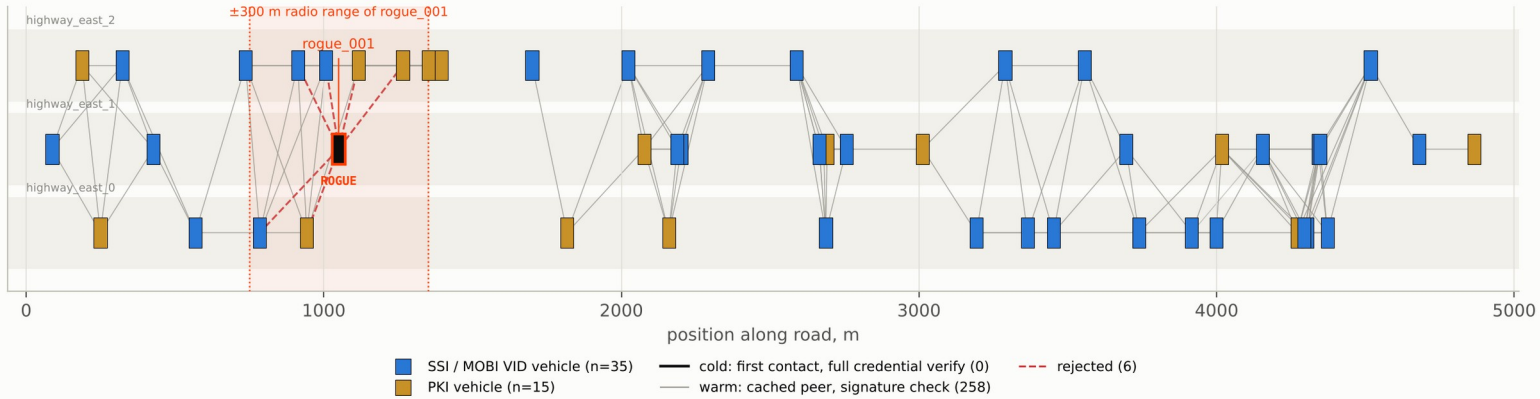
30 / 30 runs in the register.

REJECTED

AN ATTACKER ON THE ROAD

THE ROAD AT T = 12.0 S

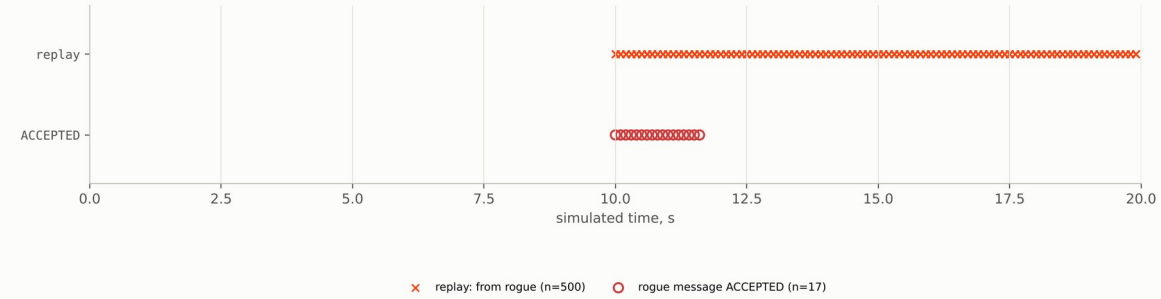
One 100 ms step: 52 signed BSM broadcasts, 264 receiver-side verifications, every one a real cryptographic operation.



trace sim_50v_20s_seed6_rogue_uncredentialed.jsonl.gz · mode simulate · seed 6 · 20 s · 10 Hz BSM · 300 m radius · ≤8 rx/broadcast · commit 3dddf6c · no radio channel, no MAC

REJECTED, AND WHY

Every receiver-side rejection in the run, by the verifier's reason (500 in total). Rogue mode: replay; rogue messages accepted: 17. Attack-board probes after the run are not receiver traffic and are not drawn.



trace sim_50v_20s_seed6_rogue_replay.jsonl.gz · mode simulate · seed 6 · 20 s · 10 Hz BSM · 300 m radius · ≤8 rx/broadcast · commit 3dddf6c · no radio channel, no MAC

617

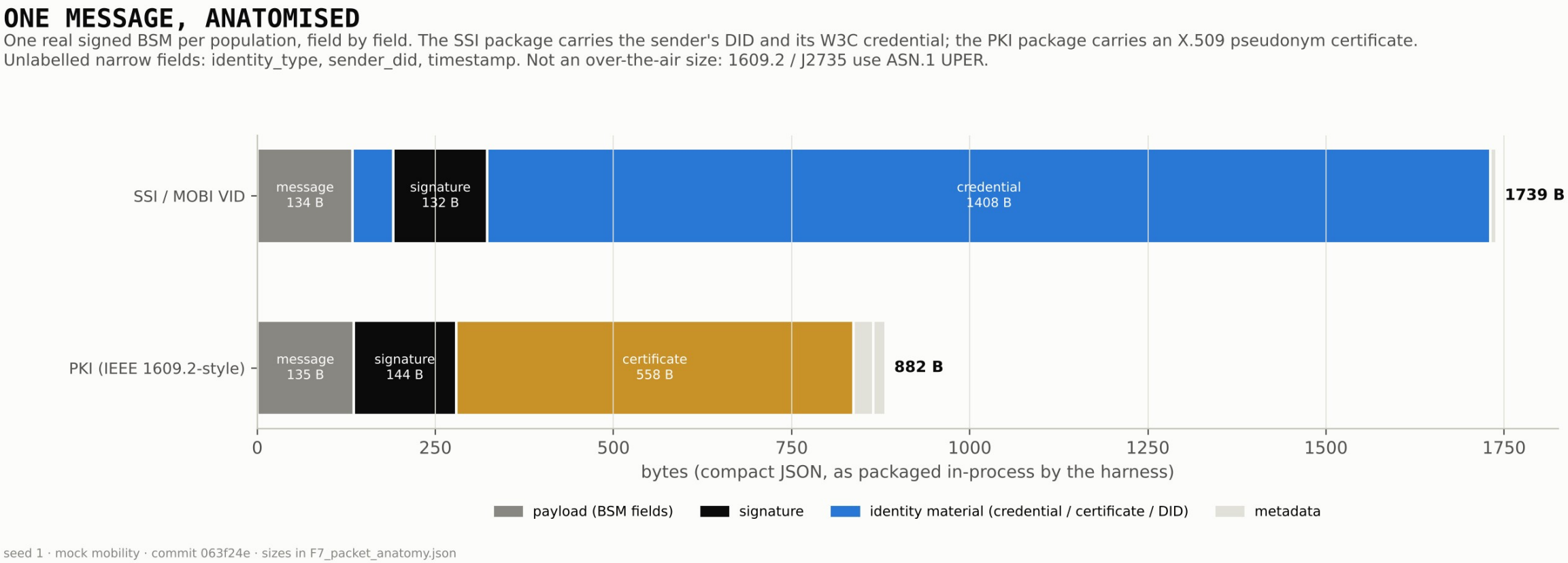
of 617 uncredentialed rogue messages rejected

17

replays accepted: fresh genuine messages relayed to receivers that missed the original

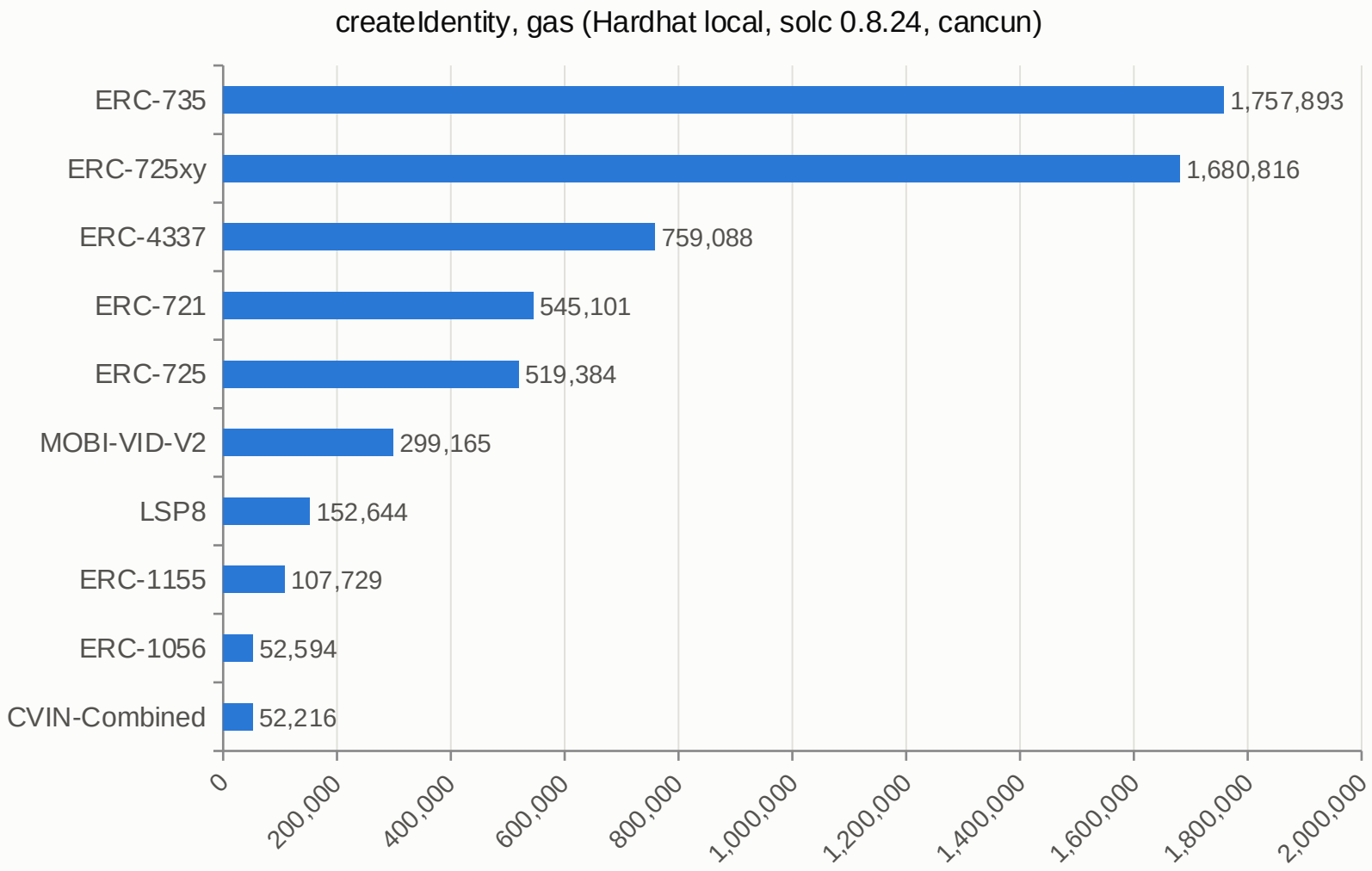
Authenticity holds; reach is extended. A position-plausibility check would close it.

ONE MESSAGE, ANATOMISED



The 1,408 B credential rides on every SSI message as the harness packages it, though a receiver needs it only at first contact. Compact-JSON sizes, not over-the-air ASN.1.

WHAT AN IDENTITY COSTS ON CHAIN



33.7×

spread, ERC-735 against CVIN-Combined

Source: 4_comparison-framework/results/gas_benchmark.json (register #25, commit 7a59058). MOBI-VID-V2 is an application profile, not a base standard.

WHAT THE PICTURES MAY CLAIM

In the pictures	Real or mock	A caption may say
Signing of each BSM	Real: ECDSA P-256 (PKI), secp256k1 EIP-191 (SSI)	each message is actually signed
Cold and warm verification	Real: X.509 chain and CRL; W3C VC and signature recovery	every link drawn is a verification that ran
Latency on any axis	Real: time.perf_counter around the cryptographic call	compute latency of identity verification only
Vehicle motion	Mock kinematic model, or SUMO 1.28 over TraCI	the mobility source, always
Radio channel	Mock: in-process delivery to the 8 nearest within 300 m	no radio, MAC or channel loss
The blockchain	Not in the V2V loop: credentials verified off chain	never a chain read per message

WHAT IS NEXT

- Decide N-4: SUMO for illustrations only, or S1–S3 at N = 30 in TraCI mode
- State the relay limit in the threat model, with a position-plausibility check as the remedy
- Send the credential on first contact, not on every message, and measure the bandwidth

